

Evidence Synthesis on Socio-Emotional Skill Returns and Training Impacts*

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December 27, 2025

Abstract

This paper synthesizes the growing evidence on the labor market returns to socio-emotional skills (SES) and the effectiveness of SES training programs. While existing studies document elevated returns to SES in the labor market, there is little consensus on which specific aspects of SES drive these gains. Utilizing established psychological measures of the Big Five Personality Traits (BFI) and taxonomy mappings from Harvard’s EASEL Lab, we define SES relevant keywords, web-scrape the literature, and construct literature database on SES returns and training impacts. DerSimonian and Laird random-effects estimates suggest that a one standard deviation increase in BFI traits is associated with a 2–3 percentage point increase in wages, with substantial heterogeneity across individual traits, demographic groups, and country income levels; assertiveness exhibits the highest returns. Training programs improve BFI traits by up to 0.5 standard deviations, with larger effects for attitude-related traits such as openness and conscientiousness. Childhood interventions and programs targeting out-of-school youth are more effective than teacher-led school interventions. In-school curricula offer scope for improvement in SES development. (150 Words)

JEL Codes: I26, J24, J31

Keywords— Socio-emotional Skills, Labor Market Returns, Skills Training Programs, Big Five Personality Traits, Quantitative Meta-Analysis

*The authors are grateful to Wendy Cunningham, Victoria Levin, Harry Patrinos, Yevgeniya Savchenko, and Jonathan Stöterau for their review of the paper. The paper benefited from comments and discussions at the World Bank Economics of Education Seminar Series, Southern Economic Association Conference, and Public Sector Economics Conference. The findings, interpretations, and conclusions expressed in this paper are entirely those of the authors and do not necessarily represent the views of the International Bank for Reconstruction and Development/World Bank and its affiliated organizations, or those of the Executive Directors of the World Bank or the governments they represent. All errors are our own.

1 Introduction

The value of skills in the labor market has shifted dramatically over the past century. In the United States, for instance, the wage gap between high-skilled and low-skilled workers has continued to widen (Tinbergen, 1974; Katz and Murphy, 1992). Skill-Biased Technical Change (SBTC) has particularly driven this trend in the late 1990s and early 2000s. The earnings of the top percentile and bottom percentile have diverged due to increasing skill bias that cannot be easily automated or off-shored (Autor et al., 2003; Acemoglu and Autor, 2011). However, with ongoing technological advancements, particularly in artificial intelligence (AI), the return to cognitive skills in the labor market has started to decline (Castex and Kogan Dechter, 2014; Deming, 2017; Ashworth et al., 2021; Hellerstein et al., 2024). This shift places specialized tasks—such as legal analysis and basic medical prescriptions—at risk (Castex and Kogan Dechter, 2014; Brynjolfsson and McAfee, 2014; Beaudry et al., 2016; Edin et al., 2022). Initially observed in advanced economies, this change has also been noted in the labor markets of other countries, albeit to varying degrees, influenced by contextual, institutional, and educational factors (Deming and Silliman, 2024).

The diminishing returns to cognitive skills have been accompanied by rising relative returns to workers’ socio-emotional characteristics. For example, Deming (2017) shows that a young worker switching to a job with 10% more social skill intensity experienced a 2.1 percentage point wage gain, whereas the wage gain from switching to a more math-intensive job declined by 0.9 percentage points in 1997 compared to 1979 (Deming, 2017). More recent evidence suggests that returns to socio-emotional skills and traits rapidly grew around the late 1990s to early 2000s, and continued growing thereafter (Edin et al., 2022). Articles commonly stress that workers with strong socio-emotional abilities sort into tasks in which they have a comparative advantage, and the increasing value of socio-emotional skills and traits raises their overall skill returns.

In light of their growing significance, scholars argue that the socio-emotional skills development is necessary to mitigate inequality in these skills. Studies find that socioeconomic disparities contribute to differences in socio-emotional skills among children (Attanasio et al., 2020), which can perpetuate a vicious cycle of wage gaps in the next generation. However, a silver lining is that socio-emotional skills and traits are malleable, allowing for gradual yet continuous improvements throughout the life cycle, influenced by life circumstances and targeted interventions (Atherton et al., 2021; Graham and Flamini, 2023).

Indeed, there is a substantial body of field experiments and quasi-experimental studies focused on effectively training socio-emotional skills. Gertler et al. (2014) found that parental interventions that included psychosocial stimulation for toddlers in Jamaica resulted in more than a 25 percent increase in earnings 20 years later. Allemand et al. (2023) demonstrated that training in conscientiousness for mature workers in Senegal’s rail construction sector led to an immediate income increase of 22.7 percent. The evidence regarding the context and timing of interventions for socio-emotional skills development and their outcomes is diverse and expanding (Gertler et al., 2014; Premand et al., 2016; Rammstedt et al., 2024).

Motivated by the growing significance of socio-emotional skills and the exponential growth of the literature, this paper collated recent empirical evidence on the wage returns to workers’ socio-emotional characteristics and on the extent to which training programs can help develop socio-emotional skills. Focusing on the Big Five Personality Traits (namely Agreeableness, Conscientiousness, Extroversion, Neuroticism and Openness), a widely used framework of socio-emotional personality traits in the literature which is particularly relevant for inferring skill education, we first synthesized their returns reported as the percentage change in a worker’s real wage from recently

published journal articles and working papers.¹ Until March 2025, there were 17,500 resources in Google Scholar with the term ‘socio-emotional skills’ in the title or abstract. Through the systematic and replicable process of Boolean keyword-based web scraping and the contents review with the assistance from Generative Pre-Training Transformer (GPT) Artificial Intelligence (AI), we narrowed them to the most relevant return estimates. There were 1,896 coefficients on the returns to cognitive and socio-emotional skills with a few foundational papers released before. Among them, we employed 613 coefficients from 16 papers for our analysis.

Our findings on the returns to workers’ socio-emotional characteristics reveal that a one standard deviation (SD) increase in each of the Big Five Personality Traits results in 2 to 3 percentage points increase in wages, which is about one third compared to a 10% return to an additional year of schooling [Patrinos and Psacharopoulos \(2020b\)](#). However, there is notable heterogeneity across the Big Five traits. Assertiveness, linked to disagreeableness, yields the highest return, with a 3.1% increase in wages, followed by Emotional Stability and Openness, each with returns above 2%, while Conscientiousness and Extroversion contribute more modestly. Beyond the main findings, heterogeneity emerges across educational attainment, gender, and country income levels, with the highest returns in high-income countries, with the caveat that most results available in developing countries are from Latin America, and sample sizes are limited to be conclusive.

Next, we investigated the effectiveness of training programs aimed at improving socio-emotional skills, focusing on whether these skills can be developed through structured interventions and identifying which program designs are most effective. Leveraging the rich meta-analytic literature on socio-emotional and non-cognitive skill training, this analysis began extracting 3,194 papers to compare the impacts of various program characteristics and narrowed them to 29 studies. The section addressed three key questions: (1) Can socio-emotional skills be developed through training interventions? (2) Is training conducive to changing the Big Five Personality Traits? (3) What are the important considerations for educational and training institutions when designing interventions to maximize results?

Our results on the impacts of training for socio-emotional skills show that interventions can improve these skills by up to half a standard deviation. Training programs targeting traits related to attitudes, such as openness (0.65) and conscientiousness (0.40), were more effective than those focused on personality, relationships, and emotions, such as extroversion (0.38), emotional stability (0.37), and disagreeableness (0.16). The latter may require longer-term interventions to achieve meaningful change. Training at earlier ages, when skills are more malleable, proved to be more effective. Out-of-school youth, who face real-world challenges, also benefited more from socio-emotional skills training. These two groups—youth and out-of-school youth—could be effective targets for socio-emotional skills development programs. Meanwhile, post-secondary education may need to more actively integrate socio-emotional skills training and assessment into the curriculum as the current returns are smaller.

The study findings also highlight that direct instruction and inquiry-based pedagogy were particularly effective, while mentor-based, collaborative, or game-based learning methods yielded less impact. Interestingly, training provided by non-teaching staff was more effective than training by regular teachers, suggesting that educators may not always be equipped with the necessary strategies to foster socio-emotional skills. This indicates a need for teachers to place greater explicit emphasis on and invest more effort into developing their students’ socio-emotional skills in

¹To avoid bias from a dependent variables conversion, we restricted our analysis to log real wage estimates. Our study excluded literature evidence employing nominal wage or other adjacent definitions as a dependent variable.

teaching through specialized programs.

This paper is organized as follows: section 2 explores the main concepts and terms in the literature to understand, classify, and measure socio-emotional skills and traits. Section 3 details the existing measures for assessing socio-emotional skills, labor market returns, and program interventions. In this section, we also describe the measures adopted in our study. Section 4 explains our human-machine collaboration process for constructing the literature database, and section 5 discusses the estimation strategy for conducting the meta-analysis. Section 6 presents our meta-analysis evidence and results on the returns to socio-emotional skills and training impacts for the skills. Finally, section 7 summarizes the key insights gleaned from our research in terms of the value added through evidence-based policy recommendations for how beneficial socio-emotional skills can be and how to introduce them as learning outcomes in curriculum and outlines potential avenues for future inquiry.

2 Skills Taxonomy and Non-Cognitive Skills

As highlighted in Heckman (2006), the structure of the brain and the process of skills development are shaped by the interplay between genetics and personal experiences. Skills are often understood as ‘a special ability to do something coming from training, experience, or practice’ that enables a person to perform specific tasks or activities effectively. On the other hand, traits are inherent characteristics or qualities that define an individual’s personality (John et al., 2008). Both skills and traits are important determinants of a person’s socio-emotional characteristics. Collectively, cognitive and socio-emotional abilities are interconnected and contribute to educational and labor market success, determined by the market’s supply and demand for these abilities.

Due to varying definitions, the skills taxonomy differs across disciplines and entities. Education scholars often categorize skills into cognitive, linguistic, and socio-emotional competencies (Heckman, 2006; Hilton and Pellegrino, 2012). These approaches often benchmark Bloom’s classification of cognitive, affective, and psychomotor learning goals (Bloom, 1974). In contrast, occupation- and task-based approaches classify skills by functionality. For instance, the content model of the United States Occupational Information Network (ONET) divides basic skills into static content and procedural skills. Using open systems theory as a benchmark (Katz and Kahn, 1978), the ONET further classifies more advanced cross-functional skills into problem-solving, resource management, social, systems, and technical skills (Peterson et al., 2001). Lastly, policy entities advocate for more structured and staged skills frameworks. Our approaches regard skills as key determinants of human development, productivity, and ultimately economic growth (Cunningham and Villaseñor, 2016). Foundational cognitive skills include basic literacy² and numeracy, which are essential for acquiring more advanced skills. Additionally, socio-emotional skills — critical to managing and regulating relationships, emotions, and attitudes — are also regarded as foundational as they are crucial for future learning and mastering higher-level skills. At a more sophisticated stage, higher-order cognitive skills encompass problem-solving, communication, information analysis, and critical thinking. The final category includes occupation-specific skills, comprising both technical and cognitive abilities tailored to specific job roles, with specialized skills like entrepreneurship also playing a significant role.

With recent evidence of the declining returns to cognitive skills in labor markets, especially in developed economies (Castex and Kogan Dechter, 2014; Brynjolfsson and McAfee, 2014; Beaudry

²These encompass both language and digital literacy.

et al., 2016; Edin et al., 2022), scholarly investigation and policy efforts are shifting toward the non-cognitive side of skills. The definition of socio-emotional skills is an evolving literature and there is not yet consensus on how to define and measure them (Whitehurst, 2016). Socio-emotional skills are known by various terms, such as non-cognitive skills, personality traits, 21st-century skills, soft skills, and more recently, higher-order non-cognitive skills (Duckworth and Yeager, 2015; Kankaraš, 2017; Cabus et al., 2021; Deming and Silliman, 2024). Similarly, “socio-emotional skills” is often regarded as a term that encompasses complex traits with overlapping boundaries. Still, the notion is gradually evolving within the discipline of economics (Deming and Silliman, 2024), and recent studies have started to focus on a few key indicators to serve as representative measures. In the next section, we will discuss some of available taxonomies, definitions, and measurement approaches found in the literature, and provide the rationale for the one chosen for this study.

Internationally, multiple frameworks exist for classifying non-cognitive skills, each reflecting distinct measurement priorities, theoretical orientations, and policy applications. The OECD Social and Emotional Skills Framework focuses on developing lifelong competencies necessary for academic, career, and personal success, aligning closely with human capital and well-being indicators (Jones et al., 2019). Meanwhile, the Collaborative for Academic, Social, and Emotional Learning (CASEL) framework structures non-cognitive skills into five core competencies, including self-awareness, self-management, social awareness, relationship skills, and responsible decision-making, primarily for application in education systems and youth development programs. These frameworks contrast with trait-based models such as the Big Five Personality Traits, which measure stable personality characteristics rather than trainable competencies. However, despite this difference, some Big Five traits can be mapped onto trainable competencies in skill-based frameworks, such as conscientiousness aligning with goal-setting and self-discipline or agreeableness relating to conflict resolution and teamwork. The literature evaluating labor market returns to non-cognitive skills has been more extensive and consistent for the Big Five framework than for other models, largely due to its longer history in psychological research and the availability of standardized measures across countries. While this is not ideal, the Big Five provides the most comprehensive evidence base for analyzing economic returns, which is why we adopt this framework in our study.

In the next section, we discuss some of the measures for non-cognitive skills that are widely accepted in the economic literature, and introduce the measures we investigate in this study.

3 Measuring Socio-Emotional Skills, Returns, and Training Impacts

3.1 Available Frameworks to Assess Socio-emotional Skills

Connected to the notion of human capital theory, skills development has long been measured by educational attainment (Card, 1999, 2001; Psacharopoulos and Patrinos, 2018). The predominant focus in economic literature has been on using years of schooling to analyze skill levels and labor market returns for workers. However, criticism has arisen regarding the reliance on years of schooling as a measure, mainly due to variations in educational quality and the environment across countries and regions. Additionally, many skills are acquired outside formal educational systems (Hanushek and Woessmann, 2008; Cunningham and Villaseñor, 2016). Furthermore, using educational attainment as a measure is somewhat crude, as it fails to distinguish between cognitive and non-cognitive aspects of skills. In response to these limitations, new metrics such as Learning-

Adjusted Years of Schooling (LAYS) have been developed to better reflect actual learning outcomes rather than mere time spent in school (Filmer et al., 2020). These measures have been incorporated into the World Bank’s Human Capital Index, helping shift the global focus from schooling quantity to education quality (skills gained) as a key driver of economic development (Hanushek, 2013).

As the analysis of the impact of skills on wages becomes increasingly important, statistical agencies are making efforts to publish skills and their competency scales relevant to occupational tasks. A growing body of literature examines skills as vectors that constitute tasks and occupations. Research on labor migration, tasks, and occupations frequently utilizes these measures (Yamaguchi, 2012; Autor and Handel, 2013; Robinson, 2018; Acemoglu and Restrepo, 2020). However, the classification of skills and the composition of skill sets for each task are not well harmonized across National Statistical Offices (NSO) and data from other sources (e.g. LinkedIn). This lack of standardization introduces subjectivity when constructing task-skill vectors. From the perspective of non-cognitive skills, the measures tend to be more occupation- and task-centric, often failing to capture unique features of non-cognitive skills, such as their latent development (Lavado et al., 2021).

To standardize the measurement of skills, scholars have turned to widely recognized assessment tools. For example, task- and test-based assessments like the Programme for International Student Assessment (PISA) and the Survey of Adult Skills (PIAAC) provide generalizable metrics for basic skills such as literacy, numeracy, and digital skills. While these assessments excel in measuring and comparing cognitive skills using task-based robust psychometric tools, they fall short in measuring the socio-emotional factors that influence non-cognitive skills and worker productivity, which are often collected through self-reported questionnaires.

Structural estimation efforts represent another approach to measuring skills. The structural approach views skills as latent variables rather than directly observable ones (Keane and Wolpin, 1997; Sarzosa and Urzúa, 2016), acknowledging that observed skills data may be particularly prone to measurement error. Researchers employing this approach use structural estimation methods to interpret survey results and test scores as outcomes of the latent skills they estimate through their models. Until recently, most studies using the structural approach have focused primarily on learning skills, such as reading comprehension (Carneiro et al., 2003). Non-cognitive traits, like extroversion or openness, have not been well quantified, despite some structural evidence of their interplay with learning skills (Acosta et al., 2020). A new approach suggested by Deming and Silliman (2024) proposes that non-cognitive or “higher-order” skills can be integrated into the human capital model as a component of task choice. This theoretical framework is gradually incorporating non-cognitive dimensions.

Lastly, the Big Five Personality Inventory constitutes one of the most widely used approaches to measure non-cognitive skills. It is a personality measure gradually developed since early 1900s until 1980s. Its five traits are agreeableness, conscientiousness, extroversion, neuroticism, and openness, and the keywords we mapped to each trait in Table 1 describe each personality more clearly. These non-cognitive skills often pertain to the management of attitudes, relations, emotions and also to high-order cognitive skills such as critical thinking and problem solving.

Many studies employ the Big Five Personality Inventory (John et al., 1991) as their primary proxy for non-cognitive skills and examine if the skills and the skill growth reflected in five personality traits, measured by scores, yields better earnings. Because the indicators show fundamental soft skills (e.g. agreeableness) for more higher-order soft skills (e.g. conflict resolution) and are measured as scores, these can be one of suitable measures of the impact of educational trainings.

Table 1: Mapping of Non-Cognitive Skills to Relevant Domains and Key Words

Big Five Traits	Related Domain	Key Words
Neuroticism	Management of attitudes	Emotional regulation, Stress management, Mindfulness
Conscientiousness	Management of attitudes	Self-control, Self-discipline, Goal-setting, Impulsivity, Diligence
Agreeableness	Management of emotions	Assertiveness, Conflict resolution, Empathy, Bystander Intervention
Openness	Higher-order cognitive skills	Creative thinking, Flexibility, Adaptability, Resilience, Problem solving
Extroversion	Management of relationships	Social skills, Communication skills, Collaboration, Teamwork

Despite this, there are number of challenges in measuring non-cognitive skills with the indicators. First, the Big Five Traits are typically measured using self-reported data, which introduces subjectivity and the potential for social desirability bias (see the Big Five Personality Traits Questionnaire in Figure A6). Respondents may unintentionally exaggerate desirable traits like conscientiousness or downplay less favorable traits such as neuroticism (Schmitt et al., 2007). Additionally, situational factors, such as mood at the time of assessment or cultural influences, can affect responses, introducing variability that may not accurately reflect stable personality traits (Podsakoff et al., 2003; Schmitt et al., 2007). This subjectivity raises the risk of measurement error, potentially biasing estimates in either direction (Almlund et al., 2011). This methodological challenge introduces biases in the measurement of non-cognitive skills and should be considered when interpreting the evidence discussed in this paper.

These limitations imply that while findings on non-cognitive skills and economic outcomes offer valuable insights, they should be interpreted cautiously. Measurement error in non-cognitive assessments can weaken causal interpretations and lead to attenuated or inflated effect sizes, affecting the precision of these estimates (Heckman and Kautz, 2012). However, the limitations do not invalidate the insights generated by such analyses; rather, they highlight the importance of triangulating findings with alternative assessment methods, such as longitudinal behavioral assessments or observer ratings, where feasible. While these methods can yield more robust and objective measures of personality traits (Abrahams et al., 2019), they are often costly and less scalable. In the context of the Big Five, recent structural estimation approaches as used by Deming and Silliman (2024), offer promising alternatives by integrating non-cognitive traits as latent factors influencing task choice, potentially minimizing the bias from subjective self-reports. However, there are not yet enough studies employing this method to provide multiple observations necessary for a meta-analysis.

Many studies also supplement or validate the Big Five traits with additional characteristics. For instance, the HEXACO Personal Inventory expands on the Big Five by including the H-Factor (Lee and Ashton, 2004; Cabus et al., 2021). Other variables include decision-making (Balcar, 2016), communication (Balcar, 2016; Acosta et al., 2020), cooperation (Balcar, 2016), emotional stability (Cunningham and Villaseñor, 2016; Acosta et al., 2020), grit (Díaz et al., 2012; Danner et al., 2019; Acosta et al., 2020), hostile attribution bias (Acosta et al., 2020), locus of control (Heineck and Anger, 2010; Bühler et al., 2020a), reading proficiency (Acosta et al., 2020), reciprocity (Heineck and Anger, 2010), risk preference (Bühler et al., 2020a), patience (Bühler et al., 2020a), perseverance

(Cunningham and Villaseñor, 2016), self-efficacy (Lavado et al., 2021), self-esteem (Lavado et al., 2021), and trust (Acosta et al., 2020).

Accordingly, this meta-analysis compiles return and training impact estimates primarily based on the Big Five Personality Inventory. By standardizing the unit of analysis to represent a one SD increase in skills and collating hundreds of coefficients for each of the five personality traits, we highlight current trends in research findings related to skill returns and training impacts.

3.2 Linking Personality Traits to Workforce Skills

While the Big Five Personality Traits are often included in discussions around workforce success, it is important to distinguish them from skills. Personality traits are relatively stable dispositions or tendencies that shape how individuals think, feel, and behave across various contexts (John and Srivastava, 1999). In contrast, skills refer to learned abilities or competencies that individuals can develop and improve over time, often through training or experience. Whereas traits are descriptive and predictive, skills are actionable and teachable. This distinction is particularly relevant in the context of workforce development, where the goal is to build capabilities that enhance employability and productivity.

While personality traits are not themselves skills, they can be linked to specific non-cognitive skills that are valued in the workplace. The Harvard Taxonomy Project has advanced this effort by mapping non-cognitive skills across frameworks to clarify similarities and definitions, with the goal of improving translation between research, policy, and practice. For the purposes of linking non-cognitive traits to concrete competencies, we draw on two widely used frameworks: the Employability Skills Framework and the Work Ready Now! (WRN) Framework. The Employability Skills Framework was developed by the U.S. Department of Education in collaboration with workforce and education stakeholders to unify core competencies across career and technical education, adult learning, and workforce development. The WRN Framework, developed by Education Development Center (EDC), outlines standards for a curriculum that equips youth in emerging economies with transferable skills, behaviors, and attitudes needed for workplace success.

Table 2 presents a mapping of the Big Five traits to skills identified in the Employability Skills Framework and the EDC Work Ready Now! (WRN) Framework, as compiled by the Harvard EASEL Lab. As shown, conscientiousness and agreeableness have the strongest representation, with clear connections to workplace behaviors such as time management, communication, and leadership. In contrast, traits like emotional stability and extraversion are represented in only one of the two frameworks. This is not because these traits lack value in the workplace, but because workforce-specific models often prioritize behaviors that are more externally observable and directly linked to task performance. Emotional stability, in particular, is frequently associated with internal states such as stress tolerance or emotional regulation, which are sometimes viewed as falling within the domain of psychological well-being rather than job readiness. As a result, some frameworks may implicitly assume these traits under broader categories like professionalism or responsibility, but do not explicitly define or target them as standalone competencies. However, emotional stability and extraversion are more fully addressed in social-emotional learning and holistic development frameworks (e.g., CASEL, OECD’s Social and Emotional Skills Framework, the ACT Holistic Framework), which include constructs such as emotion regulation, resilience, optimism, and sociability, indicating that these traits are indeed considered teachable and important, even if not emphasized in workforce-oriented models.

Table 2: Mapping of Big Five Traits to EDC Work Ready and Employability Skills

Big Five	EDC Work Ready	Employability Skills
Conscientiousness	work habits and conduct, worker and employer rights and responsibilities	demonstrates integrity, manages resources, manages time, plans/organizes
Openness	–	thinks creatively, demonstrates a willingness to learn, improves systems
Emotional Stability	safety and health at work	–
Extraversion	–	displays a positive attitude and sense of self-worth
Agreeableness	leadership, interpersonal communications, worker and employer rights and responsibilities	understands teamwork and works with others, respects individual differences, demonstrates responsibility and self-discipline, understands and uses systems, exercises leadership, responds to customer needs, manages time, negotiates to resolve conflict, demonstrates integrity, demonstrates professionalism, takes initiative, manages personnel, communicates, communicates verbally, listens actively

3.3 Labor Returns: Wage, Employment, and Beyond

In this paper, we primarily employed the monthly wage measure to investigate the returns to non-cognitive skills. Imposing restrict sample restriction for better standardization, we minimized the variation coming from measurement. In particular, we only employed log real wage in our analysis. This is because converting different wage units such as nominal wage in a country’s currency can cause bias when converting its time and currency values. Moreover, the volume of evidence on log real wages is sufficient to draw robust inferences for a meta-analysis without introducing additional measurement error from purchasing power parity corrections. Therefore, this work chose to build the evidence from the narrowest outcome variable which minimizes the bias and sets the cornerstone for broader analysis.

Wage outcomes may not fully capture the value of non-cognitive skills, which can also affect other labor market outcomes. Considering other indicators, employment cannot be overlooked (Adoho et al., 2014; Ibarraran et al., 2014; Cunningham and Villaseñor, 2016; Lavado et al., 2021; Schultheiss and Backes-Gellner, 2023). From the simple measure of comparing the employment rate of workers with varying non-cognitive skills, the job landing among unemployed trainees and the likelihood of creating business among entrepreneurs are all in this category. Other indicators the recent studies employed are career advancement (Adhvaryu et al., 2023), contract type (Cunningham and Villaseñor, 2016), educational investment (Acosta et al., 2020; Lavado et al., 2021), job satisfaction (Danner et al., 2019), occupational choice (Lavado et al., 2021), saving and investment behavior (Adoho et al., 2014), skill requirement change for job (Allen et al., 2020), tertiary educational attainment (Acosta et al., 2020), type of occupation (Cunningham and Villaseñor, 2016; Acosta et al., 2020). Studies on developing economies also investigated the formality of jobs or

other job quality indicators as the labor market is not as much institutionalized as in advanced economies (Ibarraran et al., 2014; Cunningham and Villaseñor, 2016; Acosta et al., 2020). Future research could systematically evaluate these additional labor market effects.

3.4 Classifying Training Interventions

While socio-emotional skills, like cognitive skills, have often been viewed as less malleable as individuals mature, emerging research indicates that these traits can undergo small but consistent changes throughout the life cycle, influenced by life circumstances and skills development interventions (Atherton et al., 2021). Indeed, studies show that socio-emotional skills training can be highly effective. Depending on the life stage, the most effective programs vary in their implementation forms and target audiences.

Intervention During Childhood. During early childhood, interventions that focus on play-based learning, teacher training, and parental engagement are particularly effective. Programs like Promoting Alternative Thinking Strategies (PATHS) have been found to effectively reduce problem behaviors and improve social and academic behaviors by utilizing interactive teaching strategies to foster skills, such as emotional regulation and cooperation through role-play and group activities (Domitrovich et al., 2007; Group, 1999; Bierman et al., 2010). Additionally, parental involvement in reinforcing these skills at home has been shown to accelerate child development, though it is often more challenging in low-resource environments (Luthar and Eisenberg, 2017). Teacher training, combined with structured curricula, ensures that socio-emotional skills are consistently practiced in educational settings, leading to long-term benefits that extend into adolescence and adulthood. In addition to structured interventions, the use of professional development programs like Social Emotional Engagement-Knowledge & Skills-Early Childhood provides educators with the skills necessary to foster children’s socio-emotional development. These programs have shown positive outcomes in enhancing social-communication skills in children, particularly those with challenges in this area (Siller et al., 2024).

Intervention for Adolescents and Adults. Adolescence is a critical period for shaping socio-emotional skills that are strongly linked to long-term socio-economic success. As children transition into adolescence, the focus of interventions shifts towards cognitive and social skills training. For instance, cognitive behavioral therapy (CBT) and social skills training have been used to improve self-control, communication, and resilience, particularly in at-risk youth. These interventions are most effective when paired with secure, supportive environments that mimic successful family structures, allowing adolescents to develop these skills within safe boundaries (Kautz et al., 2014).

However, several challenges persist in designing and implementing these programs. One key issue is the variability in individual responses to interventions. Factors such as age, cultural background, and personal circumstances significantly influence the effectiveness of socio-emotional skill training. Interventions must account for these differences to ensure their applicability across diverse populations, particularly in settings where cultural norms may affect how traits like conscientiousness or agreeableness are developed and expressed (Laajaj et al., 2019).

Despite these challenges, there are significant opportunities for enhancing socio-emotional skills through well-designed interventions. Research has consistently demonstrated strong links between traits such as self-control, cooperation, and emotional regulation and positive life outcomes, including academic success, career achievement, and social engagement (John et al., 2020; Stahlmann

et al., 2024). Programs that target these traits during adolescence and early adulthood can produce lasting benefits. Moreover, key life transitions, such as moving from school to work, entering leadership roles, or retirement—offer unique opportunities for targeted interventions (Kornadt et al., 2019). For example, leadership training during the transition to higher education or the workforce can enhance traits like openness and extroversion, leading to improved performance and social outcomes.

Given the malleability of socio-emotional skills and the opportunities to enhance them across various stages of life, targeted training programs offer a promising avenue for development. This study focuses on evaluating the effectiveness of such interventions, particularly those designed to improve the Big Five Personality Traits during adolescence and early adulthood. By synthesizing findings from existing meta-analyses, we aim to identify patterns and examine heterogeneity across key moderators, such as school grade, intervention setting, instructor characteristics, target population, program duration, and evaluation time frame. This approach not only leverages a vast body of evidence efficiently but also provides a nuanced understanding of how various factors influence program outcomes. Focusing on school- and work-based interventions, we have compiled evidence to form a comprehensive literature database of existing meta-analyses. The next section outlines the methodologies and processes used to construct the meta-analytic literature database.

4 Database Construction

This study employs a researcher-machine collaborative approach to create a literature database on the returns to skills and the training impacts. With the Boolean keywords web-scraping of literature repositories such as Google Scholar, Psychology and Behavioral Science Collection, and Academic Search Complete, we begin the systematic literature investigation with 1,896 skill returns and 364 training impacts coefficients and standard errors in 161 papers published between 2019 and 2024. Training a Generative Pre-trained Transformer (GPT) for information extraction, and developing reproducible code to collate coefficients, and lastly through rigorous human screening, we narrow the thousands of coefficients and standard errors into a closely harmonized database offering consistent inferences.

For the wage returns coefficients, the study reduced the initial dataset into 16 papers with 613 effect sizes, including 568 for socio-emotional and 45 for cognitive skills. It is important to note that the narrow set of literature and data can still have limitations and may not fully represent what broader literature discusses regarding the wage effects of socio-emotional skills beyond the Big 5 personality traits. However, we prioritized achieving data harmonization and alignment to the analysis topic to draw better implications on the wage returns (and training impacts) to the Big Five Personality Traits.

To estimate the impact of training programs, we conducted a meta-meta-analysis, focusing on existing meta-analyses rather than individual studies. This approach achieved efficiency because the literature on training was even larger in volume. The approach also enabled us to leverage the rigor of existing meta-analysis in screening training literature. By synthesizing meta-analyses, we identified patterns and addressed variability in factors such as program duration, setting, and target populations. To extract as many meta-analysis studies on the topic to begin the review with, we utilized several databases as sources for our analysis, including Google Scholar, the Psychology and Behavioral Sciences Collection, Education Research Complete, ERIC, APA PsycInfo, and Academic Search Complete. The initial search yielded 3,194 results. However, after applying

a rigorous screening process based on the inclusion and exclusion criteria, we finalized a list of 29 meta-analyses focusing on interventions designed to enhance socio-emotional skills, with 364 coefficients, including 61 overall training effects.³

We conducted the literature web-scraping on the returns to socio-emotional skills from March 26th, 2024, until May 15th, 2024, and the training impacts from September 1st, 2024, until October 31st, 2024. The detailed web-scraping, machine training, and database construction process such as inclusion and exclusion criteria are elaborated in the Appendix A.

5 Methodology

With the narrowly synthesized set of evidence, we conducted the Panel Random Effects (RE) estimation known as DerSimonian and Laird methods (DerSimonian and Laird, 2015). This approach was chosen over Fixed Effects (FE) meta-analysis methods because it allows for the effect sizes to vary across studies rather than assuming they are identical as in the FE methods. The flexibility of RE is particularly appropriate in this setting, as the studies included in the analysis differ in their sample sizes, target populations, model specifications, and the measurement scales of socio-emotional skills. While the dependent variable—log real wage—provides a percentage interpretation for all our selection of literature, variations in scaling and methodological choices across the literature necessitate an estimation method that can account for heterogeneity.

The RE model addresses potential sources of variance both within and across studies by assigning smaller weights to studies with significantly different effect sizes. As shown by the formula below, the weighting mechanism minimizes estimation bias arising from varying research structures while better representing more common components across studies. By reducing the influence of outliers, the RE method enhances the robustness of the analysis and provides a more generalized estimate of the returns to socio-emotional skills. While the DerSimonian and Laird estimator can introduce publication bias when the number of studies is limited (Guolo and Varin, 2017), the hundreds of estimates included in our analysis provide sufficient power to mitigate this concern.

$$w_i = \frac{1}{\sigma_i^2 + \tau^2} \quad (1)$$

Although the RE model was the primary estimation strategy due to its suitability for synthesizing heterogeneous evidence, we conducted robustness checks using FE meta-analysis. The FE approach served as a benchmark to confirm the consistency of results under the assumption of uniform effect sizes. Additionally, we identified the preferred model from each paper and performed another robustness check using the RE model on this narrower pool of effect sizes. However, we interpret these estimates with caution, recognizing that a limited number of effect sizes may pose challenges for the DerSimonian and Laird estimator.

To compute the pooled effect size ($\hat{\mu}_j$) for a given socio-emotional skill j , we apply the following weighted average formula:

$$\hat{\mu}_j = \frac{\sum w_{ij} ES_{ij}}{\sum w_{ij}} \quad (2)$$

³This seems like a relatively limited sample of papers for a meta-analysis but the number of estimates is sufficient for robust inference. Data are still being gathered since both measuring socio-emotional skills and tracing individuals who benefit from training after training completion are complex endeavors

where w_{ij} is the inverse-variance weight for study i related to skill j , as defined in (1), and ES_{ij} represents the effect size for study i corresponding to skill j .

6 Results

6.1 Labor Market Returns to Socio-Emotional Skills

Descriptive Statistics

Table 3 presents the descriptive statistics of the 613 coefficients from our labor returns analysis, divided into studies examining high-income countries (HICs) and those focusing on developing countries. Studies on high-income countries (HICs) account for the majority of the observations, with 359 out of 613 coefficients, while studies on developing countries contribute 254 estimates.

For both groups, more than 90% of the estimates we collated are on the Big Five personality traits with the remaining 10% on cognitive skills. Studies on developing countries have a slightly higher share (93.3%) compared to those on HICs (92.2%). The inclusion of education as a control variable varies substantially. In studies on developing countries, 73.2% of estimates include education level as controls, compared to only 28.4% in studies on HICs. This reveals a greater emphasis in developing country contexts on understanding how education interacts with skill effects, while studies on HICs may focus more on standalone skill effects. It is worth noting that most of the studies on developing countries in our sample are from Latin America; thus, caution is warranted in generalizing these results to the entire developing world.

Table 3: Descriptive Statistics, Labor Returns Analysis by Income Group

	All	High income countries	Developing countries
Number of estimates	613	359	254
Estimate type (%)			
Big Five	92.7	92.2	93.3
Cognitive	7.3	7.8	6.7
Education control (%)			
Yes	47.0	28.4	73.2
No	53.0	71.6	26.8
Gender (%)			
Female	22.7	37.3	2.0
Male	22.8	37.6	2.0
Mixed	54.5	25.1	96.1
Methodology (%)			
OLS	28.6	30.1	26.4
IV	71.4	69.9	73.6

Gender focus shows another steep differences. In studies on HICs, 37.3% and 37.6% of estimates pertain to female and male samples, respectively, with 25.1% analyzing mixed samples. In contrast, 96.1% of estimates in studies on developing countries use mixed samples, with only 2.0% focusing on female or male-specific effects.

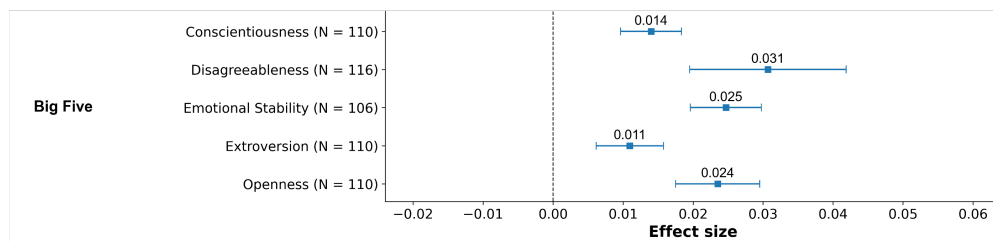
Finally, methodological differences are evident between the two groups. Studies on developing countries rely more heavily on instrumental variable (IV) methods (73.6%) compared to studies on HICs (69.9%). Meanwhile, ordinary least squares (OLS) methods are used slightly more frequently in studies on HICs (30.1%) than in those on developing countries (26.4%).

The Returns to Socio-Emotional Skills

Figure 1 shows the estimation results for the returns to socio-emotional skills with all 613 effects in our dataset. The results indicate that a one SD increase in socio-emotional skills is associated with a 2% increase in wages. While this effect is smaller compared to the returns to one year of schooling, the narrow confidence interval suggests that socio-emotional skills do yield statistically significant wage returns to workers. The findings are consistent with existing literature that highlights the generally larger wage returns to schooling, particularly in settings where educational attainment and technical expertise are prioritized (Heckman et al., 2006).⁴

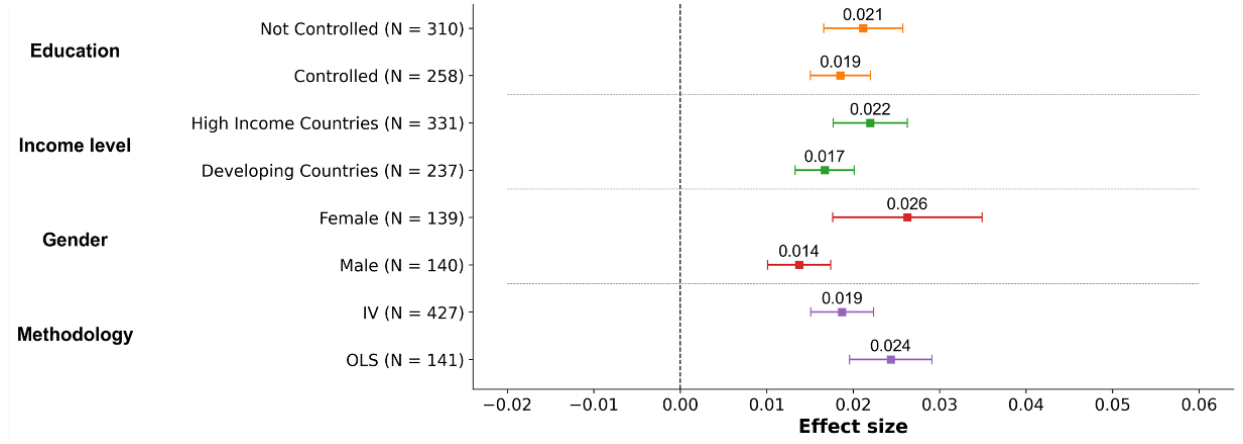
Among the Big Five personality traits, disagreeableness yields the highest returns, with a 1 SD increase associated with a 3.1% increase in wages, followed by emotional stability (2.5%) and openness (2.4%). These findings suggest that assertiveness, often associated with disagreeableness, may play a significant role in wage determination, particularly in labor markets where negotiation, leadership, or individual competitiveness is valued. The relatively high returns to openness highlight the economic value of creativity, adaptability, and intellectual curiosity in modern, dynamic work environments. Similarly, emotional stability, which reflects an individual’s ability to manage stress and remain calm under pressure, is highly valued across occupations that require resilience and effective interpersonal interactions. In contrast, conscientiousness (1.4%) and extroversion (1.1%) have lower but still meaningful returns, indicating the importance of diligence, organization, and sociability in wage outcomes.

Figure 1: The Wage Returns to Big Five Traits vs Cognitive Skills, RE, All Sample



⁴Recent studies also suggest the complementarity between cognitive and socio-emotional skills, particularly among high-skilled workers, employing skills or task data on cognitive and socio-emotional traits to investigate such complementarity.

Figure 2: Heterogeneity Effects, RE, All Sample



When comparing countries by income level, high-income countries demonstrate higher returns, with a 2.2% increase in wages per 1 SD increase in socio-emotional skills, as shown in figure 2. Developing countries show lower returns, at 1.7%. The results suggest that soft skills are slightly more valued in the labor markets of advanced economies than in developing economies. This difference may also reflect the influence of contextual factors, such as economic structures and cultural norms.

Gender is an important determinant of the returns to socio-emotional skills. The estimated average effect size is higher for women (2.6%) than men (1.4%), and the specific Big Five traits that contribute most to wage increases vary by gender. For women, disagreeableness is associated with the highest returns, with a 7% wage increase per 1 SD increase. This finding aligns with the hypothesis that assertiveness, often perceived as disagreeableness, may help women navigate traditionally male-dominated labor markets and overcome gender biases in negotiations and leadership opportunities. Conversely, for men, the highest returns are observed for Openness, with a 2% increase per 1 SD increase. This could reflect the premium placed on creativity and intellectual curiosity in male-dominated industries where innovation and exploration are rewarded.

Models with and without education as a control variable yield almost similar results, with returns to a 1 SD increase in Big Five Personality Traits estimated at 2.1% when education is not controlled and 1.9% when it is included. The slightly higher returns without controlling for education likely reflect the complementarity between cognitive and socio-emotional personality traits, as schooling can indicate both cognitive and socio-emotional personality traits. For example, it enhances traits like persistence, pro-social behaviors, and emotional regulation. When education is excluded, the estimated returns to socio-emotional personality traits may partially capture the impact of cognitive skills embedded in educational attainment, slightly overstating the effect. However, the modest difference in effect sizes demonstrates that personality traits independently drive wage outcomes, reinforcing their importance in the labor market beyond what is explained by schooling.

Our estimates are robust across different sample definitions. We pooled effects sizes using only the preferred models identified in each paper. These are models expressed explicitly by authors as their main model, or models that with the most comprehensive set of control variables. The

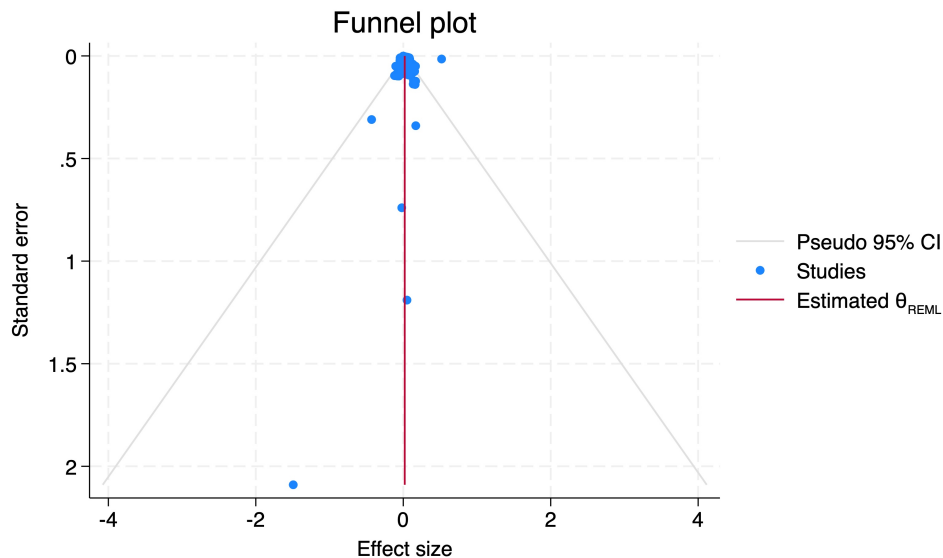
estimates restricted only to the preferred models show less precise estimates as effect sizes were significantly reduced. Figure A7 shows that the average returns to Big Five personality traits remains in the neighborhood of 2%, while for cognitive skills, the returns were lower at 6%.

We also pooled effect sizes using the fixed effects model as presented in Figure A8. The returns are slightly lower but with greater level of precision. Finally, we selected only the models that included all Big Five personality traits to ensure the capture of partial effects and to avoid potential bias that may arise from the omission of any trait. This approach allows us to estimate the overall returns to socio-emotional personality traits as the sum of the partial effects of all Big Five traits. The results presented in Figure A9 are consistent with our main findings, reinforcing the robustness of our estimates.

Assessing Publication Bias

To assess publication bias, we use the funnel plot, which is a widely used tool in meta-analysis to visually assess when studies with significant or favorable results are more likely to be published or included in an analysis than those with non-significant results. The funnel plot displays the relationship between the effect sizes of studies and their corresponding standard errors. In the absence of publication bias, the plot should resemble a symmetrical inverted funnel, with larger studies (smaller standard errors) clustering around the overall effect size at the top and smaller studies (larger standard errors) spreading symmetrically at the bottom. Asymmetry or a lack of small studies with non-significant results may indicate potential bias.

Figure 3: Returns Estimation: Publication Bias



For our sample, the funnel plot in Figure 3 shows a mostly symmetrical distribution of studies around the estimated effect size, which suggests that publication bias is minimal. Most studies are clustered near the top of the funnel, reflecting precise estimates from larger sample sizes. However, there is a slight scarcity of smaller studies with non-significant or negative effect sizes in the lower

part of the funnel, which could indicate a mild degree of publication bias or simply reflect the challenges in detecting and publishing small, non-significant effects. Additionally, a few studies are outliers, with effect sizes that deviate significantly from the cluster due to heterogeneity in study designs or sample characteristics rather than bias.

6.2 Training Interventions for Socio-Emotional Skills Development

Our estimates in Figure 4 show that training interventions aimed at developing socio-emotional skills have an average effect size of 0.46 SD. The revealed effect suggests that training interventions can be effective in developing socio-emotional skills to realize their returns. Summary statistics describing the characteristics of the corresponding estimates are provided in Table 4

However, there is significant heterogeneity in effectiveness across the Big Five personality traits (Figure 5). Interventions are most effective for openness ($\theta = 0.65$) and conscientiousness ($\theta = 0.40$), with moderate effects observed for emotional stability ($\theta = 0.37$) and extroversion ($\theta = 0.38$). In contrast, the effect size for disagreeableness ($\theta = 0.16$) is notably lower, suggesting that traits such as empathy and standing up to others may be more resistant to change through short-term training. Overall, these findings highlight that while socio-emotional skill interventions can yield significant improvements, traits like openness and conscientiousness appear to be particularly responsive to training efforts. It may be easier to train individuals to manage their attitudes than to manage relationships or emotions. Developing disagreeableness may require longer-term or more immersive interventions.

Table 4: Descriptive Statistics, Socio-emotional Skill Training Programs

	Overall	Agreeableness	Conscientiousness	Emotional Stability	Extraversion	Openness
Overall (N)	56	3	11	20	8	11
Education estimates (N)	52	2	9	5	10	26
ES (%)	23.08	50	33.33	0	20	23.08
HS (%)	9.62	50	0	0	10	11.54
Out of school (%)	7.69	0	11.11	0	10	7.69
PE (%)	13.46	0	0	0	40	11.54
PS (%)	46.15	0	55.56	100	20	46.15
Setting estimates (N)	7	0	2	0	2	3
Classroom (%)	57.14	0	50	0	50	66.67
Outside classroom (%)	42.86	0	50	0	50	33.33
Instructor type (N)	16	2	4	3	7	0
Other (%)	56.25	50	50	66.67	57.14	0
Teaching staff (%)	43.75	50	50	33.33	42.86	0
Targeting (N)	14	0	2	4	2	6
General (%)	7	0	1	2	1	3
Targeted (%)	7	0	1	2	1	3
Technology enhanced (N)	48	3	6	6	9	24
Technology-enhanced (%)	14.58	66.67	0	33.33	11.11	8.33
Traditional (%)	85.42	33.33	100	66.67	88.89	91.67
Training duration (N)	11	2	0	5	1	3
1–3 months (%)	36.36	50	0	40	0	33.33
< 1 month (%)	9.09	0	0	0	0	33.33
> 3 months (%)	54.55	50	0	60	100	33.33
Evaluation timing (N)	6	0	2	0	0	4
Follow-up (%)	50	0	50	0	0	50
Immediate (%)	50	0	50	0	0	50

Figure 4: Training Overall Impact on Big Five Personality Traits, RE, All Sample

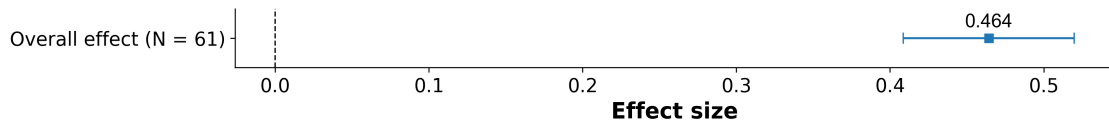
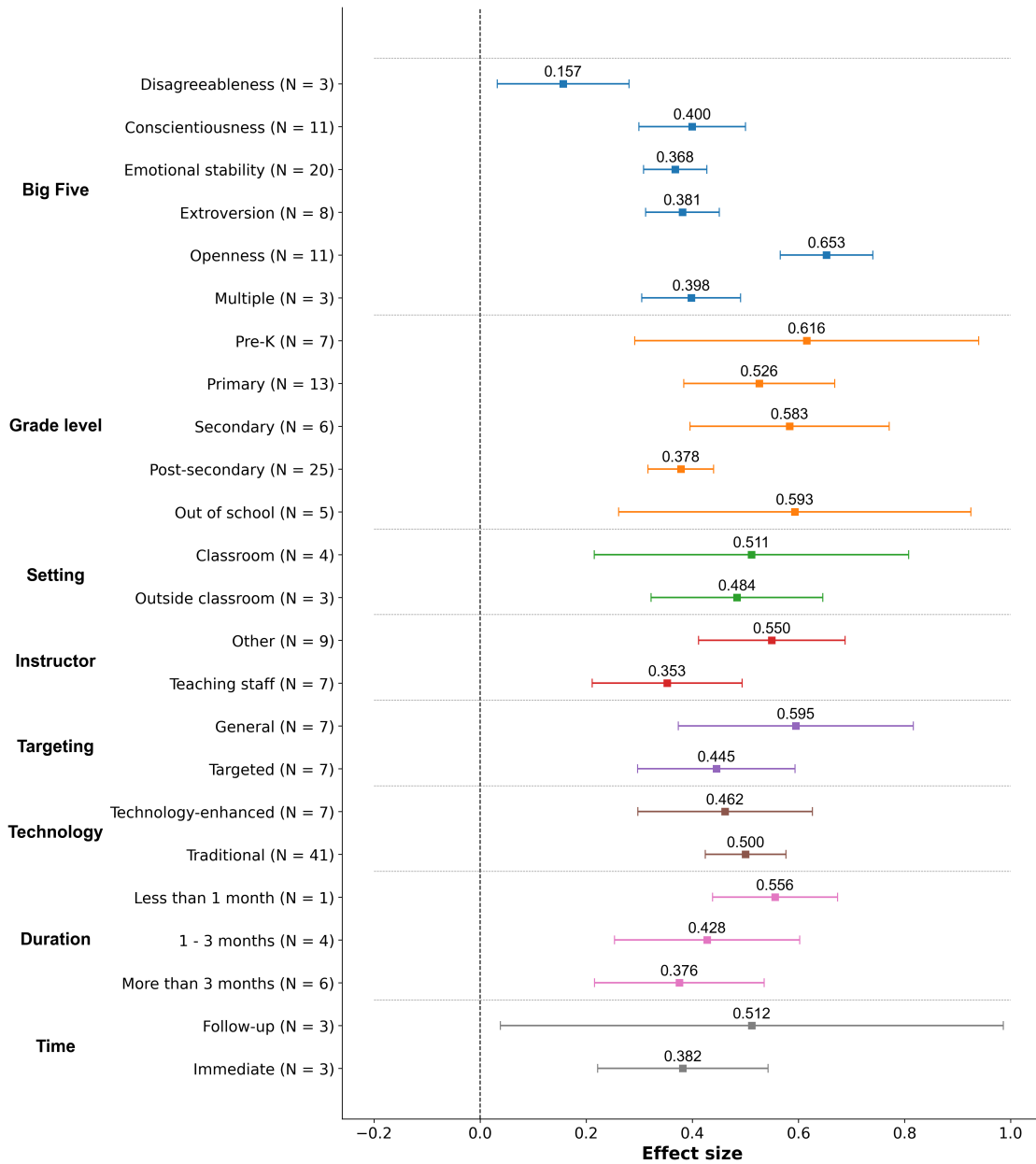


Figure 5: Heterogeneity of Training Impacts, RE, All Sample



Effects by Educational Level

The effectiveness of socio-emotional interventions varies significantly by educational level. The greatest benefits are realized for children enrolled in the primary level and for out-of-school youth, emphasizing the significance of early interventions when children’s social skills are malleable and among economically disengaged youth, who are often secondary school dropouts. For these groups, training could be a successful mechanism to develop foundational socio-emotional skills like emotional control and self-discipline (Duckworth et al., 2014). Primary ($\theta = 0.53$) and secondary school ($\theta = 0.58$) interventions also show strong positive effects, indicating that structured programs at these stages can continue to refine and enhance personality traits as children develop.

At the post-secondary level, however, the impact is lower ($\theta = 0.38$). While socio-emotional skills remain malleable throughout life, several factors may contribute to reduced effectiveness at this stage. Evaluations at the post-secondary level tend to focus more on cognitive skills such as critical thinking, technical expertise, and subject knowledge. Academic success is largely determined by grades, exams, and mastery of content, which prioritize cognitive development over socio-emotional traits. socio-emotional skills like perseverance, adaptability, or emotional regulation are important but often operate in the background, supporting cognitive skill acquisition rather than being directly evaluated.

There is an interesting contrast between the impact of socio-emotional skills training interventions at the post-secondary level and for individuals who are already out of school. While the impact of such training for post-secondary students is relatively low, the effect is notably higher for those who are no longer in formal education ($\theta = 0.59$). This higher impact for out-of-school youth can be attributed to several factors. First, they may face more immediate real-world challenges, where socio-emotional skills are crucial for navigating work environments, career transitions, and personal responsibilities. The relevance of these skills in daily life increases their engagement with training programs, leading to more significant gains. Additionally, without the structured environment of formal education, out-of-school individuals must rely more on their own ability to manage time and tasks, making socio-emotional skills training even more impactful. The practical, immediate application of these skills in diverse real-world contexts further enhances the effectiveness of interventions.

This disconnect suggests a need for post-secondary institutions to rethink how they approach the development of socio-emotional skills. Currently, academic performance metrics focus on cognitive competencies, such as exams, essays, and projects, which reward knowledge retention and critical thinking but often fail to address the importance of socio-emotional skills. To address this, colleges should better integrate socio-emotional skill development into their curricula and reward systems. Offering more experiential learning opportunities, such as group projects and leadership roles, can provide avenues for students to practice and demonstrate these skills. Additionally, adopting holistic evaluation systems that recognize communication, problem-solving, and collaboration would incentivize students to invest in their socio-emotional skill development.

Effects by Program Setting and Instructor

The setting in which socio-emotional skills are taught appears to have a minimal influence on training effectiveness, with both classroom-based programs ($\theta = 0.51$) and those conducted outside the classroom ($\theta = 0.48$) yielding moderate, significant effects. This suggests that the physical environment—whether in a traditional classroom or an alternative setting—does not drastically af-

fect outcomes. Likewise, technology-enhanced programs ($\theta = 0.46$) perform similarly to traditional in-person approaches ($\theta = 0.50$), indicating that socio-emotional skills can be developed effectively regardless of whether the training is delivered face-to-face or through digital platforms.

However, the instructor played a significant role in training effectiveness. Surprisingly, regular teaching staff, such as teachers, were less effective at training socio-emotional skills compared to others, such as nonprofit trainers, psychologists, and life coaches. These findings suggest that teaching staff often lack the consideration, expertise, or strategies needed to effectively develop students' or trainees' soft skills. It appears that regular educators need to place greater emphasis on socio-emotional skills and develop more effective approaches for teaching them.

Effects by Curricula and Pedagogy

While it is uncommon in the meta-analytic literature to explore the impact of training pedagogy, there is some evidence that it plays a significant role in determining the effectiveness of socio-emotional skill interventions. The heterogeneity observed across different pedagogical methods suggests that the choice of instructional approach can substantially influence outcomes. For example, Zhang et al. (2023) found that direct instruction ($\theta = 0.822$) was one of the most effective approaches for socio-emotional learning in virtual reality programs, even though it was one of the least utilized methods. This highlights the potential of direct, structured teaching to produce significant skill development when applied correctly. Additionally, inquiry-discovery methods ($\theta = 0.753$) also emerged as highly effective, fostering creativity and problem-solving by encouraging exploration and active engagement. Wang et al. (2024) similarly emphasize that inquiry-based learning is particularly useful in promoting divergent thinking and creativity. Meanwhile, mentor-based ($\theta = 0.456$) and collaborative learning ($\theta = 0.438$) methods demonstrated moderate effectiveness, highlighting the importance of personalized guidance and teamwork in skill development. In contrast, methods such as game-based learning ($\theta = 0.26$) exhibited lower impacts, though they may still enhance engagement and motivation.⁵

Effects by Targeting Approach

General interventions ($\theta = 0.60$) tend to be more impactful than targeted programs ($\theta = 0.45$), but both approaches have vital roles in designing effective socio-emotional skills training interventions. While broader applicability often leads to higher, more consistent outcomes across diverse populations in general programs, making them ideal for large-scale interventions, targeted programs are indispensable for meeting the specific needs of certain high-risk groups. Universal programs may be more straightforward because they do not focus solely on individuals who may present more complex challenges and require addressing a range of external constraints such as socioeconomic challenges, disabilities, and family dynamics, which can limit the effectiveness of training (Piquero, et al, 2016).

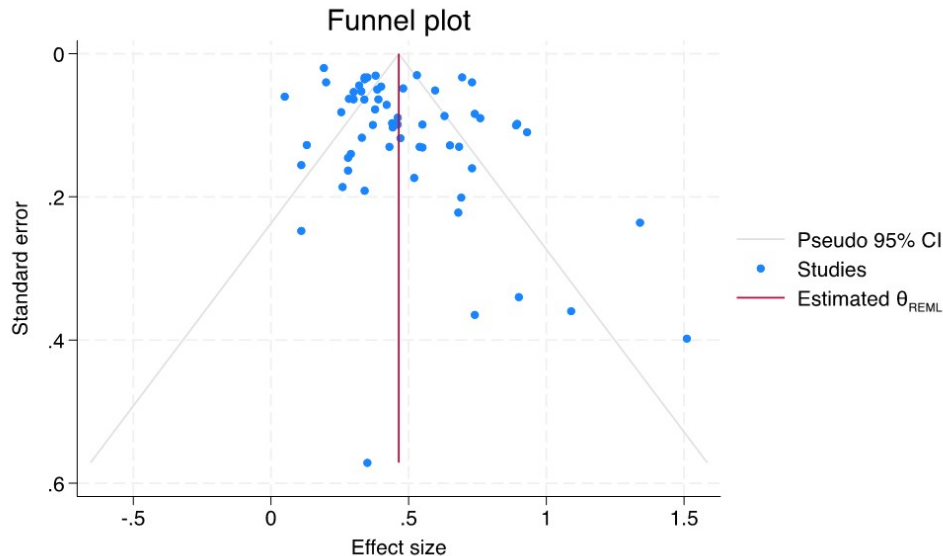
However, it is crucial to emphasize that this finding does not discourage the use of targeted interventions; rather, it highlights the need for thoughtful consideration in their delivery to enhance effectiveness. Targeted programs, while sometimes facing challenges due to these external factors, are essential for addressing specific traits or the needs of particular groups. They allow for

⁵It is important to note the higher effectiveness of specialized staff such as psychologists or life coaches as opposed to regular teachers in delivering non-cognitive skills training. This points to the importance to either offer additional training to teachers to deliver the relevant socio-emotional skills modules or to incorporate such specialized staff

deeper, more focused improvements in socio-emotional skills, making them indispensable for specialized populations. Therefore, designing interventions that consider how to mitigate the dilution of focus and maximize impact, particularly in environments with numerous external challenges, is key. Combining general and targeted approaches strategically ensures that interventions are both broadly impactful and finely tuned to meet specialized needs.

The funnel plot Figure 6 shows some minimal publication bias, potentially due to publication bias from each meta-analysis being carried over.

Figure 6: Training Estimation: Publication Bias



7 Conclusion and Policy Implications

In this study, we conducted a systematic literature review on recent empirical findings regarding the wage returns of the Big Five personality traits and the impacts of training on socio-emotional skills development. With the growing importance of non-cognitive skills for success in labor markets, evidence on these skills has expanded significantly over the past decade or two.

By constructing a highly synthesized database of relevant studies using a human-machine collaboration involving web scraping and a large-scale AI language model (GPT), we performed a DerSimonian and Laird Panel Random Effects estimation. The highly reproducible and efficient database construction process—including 1) literature repository web scraping, 2) data extraction with GPT, 3) final human screening, and 4) code-based coefficient extraction—is one contribution of this paper together with the scholarly findings.

Our results yield four key takeaways. First, we found that a one standard deviation (SD) increase in each of the Big Five personality traits results in an average 2 percentage points wage increase (Patrinos and Psacharopoulos, 2020b). Second, we observed that the returns to socio-emotional skills are 0.5 percentage points higher in high-income countries (HICs) compared to developing countries, suggesting that socio-emotional skills are more highly valued in advanced

economies. Since much of the evidence from developing countries comes from Latin America and the Caribbean, these findings may not fully represent the broader developing world.

Third, our results indicate that training for socio-emotional skills is more effective in the early stages of formal education and for out-of-school youth. In contrast, training in post-secondary education was less effective, with assessments in this stage still largely focused on cognitive skills. This finding suggests that socio-emotional skills are more adaptable in younger students, making early education a critical period for their development. Additionally, these skills are particularly valuable for youth as they navigate the challenges of the real world beyond the classroom. In addition, the educational approach at the post-secondary level may need revisiting, with a greater explicit emphasis on socio-emotional skills.

Fourth, we found that non-teaching staff—such as nonprofit trainers, life coaches, and psychologists—are more effective in training socio-emotional skills compared to regular teaching staff. This indicates a need for teachers to give more consideration and efforts to train and invest for pedagogical development and design in fostering students' socio-emotional skills.

Despite these contributions and progress, several directions for future research are suggested to fill remaining research gaps. First, examining other labor market indicators, such as employment outcomes, could provide more comprehensive insights into the role of socio-emotional skills. Second, validating our findings using alternative socio-emotional skill indicators, such as grit or cooperation, would strengthen the robustness of the results. Third, investigating the complementarity between the Big Five personality traits and cognitive skills would provide valuable insights into how these skill sets interact in shaping labor market outcomes and a better assessment of the overall rate of return of socio-emotional skills.

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A Appendix

A.1 Web-scraping in Google Scholar

We collected all research and policy publications released between 2019 and 2024 by web-scraping Google Scholar and a few more research repositories. We used the Python package Scholarly for the web-scraping, employing Boolean "Yes" or "No" operations for each keyword. Through several pilot tests, six batches of keyword searches yielded a vast pool of literature on skill returns. For the training impacts, we mapped commonly used terms associated with training for each of the Big Five personality traits—such as "creativity training" for openness—to guide our keyword selection.

We also tested several other keywords such as "Leadership", "Cooperation", "Organized", "Altruism", and "Latent Skills" to find proper supplementary measures for non-cognitive skills. When the use of keywords retrieved unrelated studies (e.g., the keyword "Cooperation" retrieved many papers about the benefits of economic cooperation in the context of bilateral trade agreements), they were excluded from our final list.

A.2 Variables Extraction by the Generative Pre-Training Transformer (GPT)

After web-scraping literature by keywords, we retrieved information from each paper and meta-analysis, including paper title, authors, publication year, journal name, journal impact factor, abstract, JEL codes if present, keywords the authors specified, number of citations, main methodology, dependent variables, non-cognitive skill indicators, other key independent variables, the estimated return coefficients, p-values and statistical significance, sample size, whether the study is on training programs, main findings, paper link, and whether the paper conducted causal analysis or not.

To enhance the efficiency of the data collection, we trained OpenAI's GPT-4.0 to extract the titles, abstracts, and other important information. GPT-4.0 barely generated errors in the simple information collecting, but there were occasional errors when reproducing complex, qualitative, and lengthy content ⁶. Therefore, we corrected machine-made errors through human screening afterward. With the trial of the machine-assisted data search, and enhanced efficiency in our information collection.

A.3 The Database Refinement

A.3.1 List of Final Papers on the Non-cognitive Skill Returns

Figure A1 describes our searching, filtering, and coding approaches for the skill returns estimation.

Boolean keyword searches were employed for web-scraping, combining terms related to wage returns, non-cognitive skills, and labor market outcomes. Google Scholar offered a credible volume of papers, so we scraped the database for the analysis and excluded all papers retrieved unrelated to the research ⁷.

⁶AI experts have suggested that the OpenAI GPT search may have errors of up to five percent. Recently developed tools, such as ImpactAI at the World Bank's Development Impact (DIME) group, and their broader application versions and algorithms, could further enhance this process once they are released.

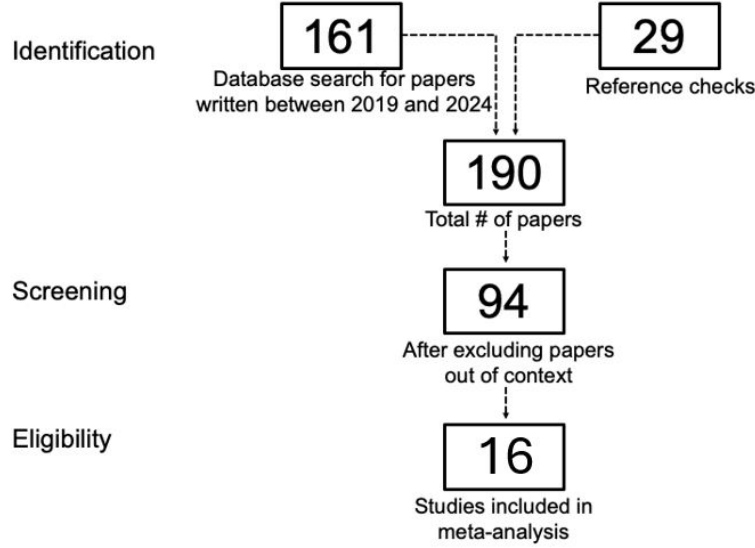
⁷Whereas, the training analysis used systematic review papers only because individual papers report impacts with large heterogeneity. With this approach, we searched more repositories to collate as many systematic review papers.

Figure A1: The Literature Selection and Analysis Process

Search strategy How did we extract the papers?	Inclusion rules Which papers should be included?	Coding rules How did we classify the papers?
- Extracted papers from Google Scholar for published works between 2019 and 2024 - Included papers found from looking up the references - Keyword searches using Boolean operator: <pre>= ("returns" OR "income" OR "wage" OR "earnings") AND ([vector of non-cognitive skill keywords]) AND ("labour market" OR "labor market" OR "employment")</pre>	- Papers written in English - Uses a continuous, standardized measure of non-cognitive skills. Papers using dummy variables to represent non-cognitive skills (e.g., training dummy) are excluded - Wage as the dependent variable - Estimation equation is expressed as semi-log equation, such that the estimated effect is in terms of % increase in wage for a 1 SD change in the non-cognitive skill - Studies from all geographic regions, industries and occupations - No restriction on the specific Big Five inventory	- Collected estimates for the main model along with heterogeneity and robustness checks - Identified the quality of evidence, particularly whether an IV is used - Identified the potential moderators: - Control for educational attainment - Control for cognitive skills - Gender - Region/continent

To ensure comparability and consistency, we refined the dataset through rigorous human screening, applying clear inclusion criteria. Papers were restricted to those written in English, reporting wage as the dependent variable, and using a continuous, standardized measure of non-cognitive skills (e.g., a 1 SD increase). Studies using dummy variables for non-cognitive skills were excluded. Additionally, the estimation equations needed to follow a semi-log form, reporting percentage changes in wages. To enhance comparability further, we restricted the analysis to studies reporting impacts in terms of SDs, a consistent metric that allows results to be compared on the same scale. This additional refinement reduced the dataset to 16 papers with 613 effect sizes, including 568 for non-cognitive skills and 45 for cognitive skills. Focusing on SD impacts simplifies interpretation and minimizes inconsistencies arising from varying measurement units or methods across studies. The PRISMA chart is shown in Figure A2, and the final list of papers is in Table A1.

Figure A2: PRISMA chart - Returns to Non-cognitive Skills



Finally, we applied coding rules to extract consistent and detailed data, such as estimates from the main models, heterogeneity checks, and robustness analyses. We classified study quality as follows: 1 = RCT/Experimental design, 2 = Quasi-experimental (DiD, RDD, IV, Matching, etc.), 3 = Correlational evidence, 4 = Qualitative evidence. Figure A3 shows the hierarchy of rigor we employed. Also, we identified potential moderators like educational attainment, cognitive skill controls, gender, and geographic region.

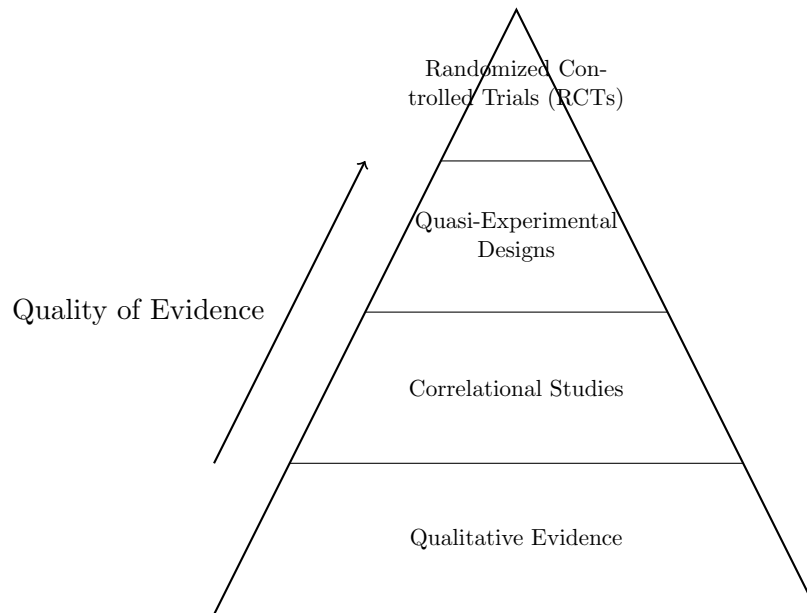
With the merit of hundreds of observations, we could subsample the evidence by education control, gender, and country income class. We employed the 2024 World Bank Group Country Classification by Income for the countries grouping. The indicator classifies the countries into ‘Low-Income,’ ‘Low-Middle Income,’ ‘Upper-Middle Income,’ and ‘High-Income’ countries⁸. However, given the limited sample size for each group, we eventually used only two income groups - High-Income, representing developed countries, and Developing Countries (all other income levels).

A.3.2 List of Final Papers on the Program Intervention

Figure A4 summarizes our approach to searching, filtering, and coding program intervention studies. Given the vast amount of primary evidence on training effectiveness, we focused our search by prioritizing existing meta-analyses. This approach is not only more practical, given the time and resources required to analyze thousands of individual studies, but it also allows us to draw on the comprehensive findings and identify broader patterns and address variability in factors such as program duration, setting, and target populations. To find relevant meta-analyses, we used a Boolean keyword search across multiple literature repositories. Since training programs often don’t

⁸The income group threshold is each GNI per capita of \$1,135 or less for the low-income, between \$1,136 and \$4,465 for the low-middle income, \$4,466 and \$13,845 for the upper-middle income, and above \$13,846 for the high-income countries.

Figure A3: The Evidence-Based Pyramid



explicitly reference the Big Five traits, we mapped each trait to commonly used terms in training interventions. Figure A4 shows the specific terms we used for this process.

Figure A4: Keywords mapped to each Big Five Personality Trait

Search strategy How did we extract the papers?	Inclusion rules Which papers should be included?	Coding rules How did we classify the papers?
- Mapping each Big Five Personality Trait to commonly used terms in training interventions (e.g., “creativity training” for openness, “stress management” for emotional stability) - Extracted from 5 different databases with significant collection of papers on training effectiveness - Search yielded 3,194 results - Keyword searches using Boolean operator: = “meta-analysis” AND (“self-confidence”) AND (“training” OR “intervention” OR “program”) AND (“school” OR “students” OR “adolescents”)]	- Papers written in English - Only meta-analyses are included, - Focus on school- or work-based interventions; excluded if family- or home-based - Excludes specific populations (e.g., individuals with disabilities, athletes). - Narrowed down to a final list of 29 meta-analyses	- Collected estimates for the main model along with heterogeneity estimates - Identified the potential moderators: - Grade level - Intervention setting - Instructor - Targeting - Technology use - Duration - Time frame for evaluation

We also expanded our database search to supplement our Google Scholar search with papers in academic databases, including the Psychology and Behavioral Sciences Collection, Education Research Complete, ERIC, APA PsycInfo, and Academic Search Complete. Initially, the search

yielded 3,194 results, but several were duplicates or irrelevant to the research objectives. Then, we applied strict inclusion criteria, limiting our sample to English-language meta-analyses that focused on school- or work-based training interventions. Studies examining home-based approaches or targeting specific populations, such as individuals with disabilities or athletes, were excluded. This rigorous screening process allowed us to narrow the dataset to a final list of 29 meta-analyses, which aligned with our focus on interventions designed to enhance non-cognitive skills. From these meta-analyses, we collected 364 estimates from the main models as well as heterogeneity estimates to capture variations in training effectiveness. The PRISMA chart is shown in Figure A5, and the list of papers can be found in Table A2.

Figure A5: PRISMA chart - Program Training



We further applied a structured coding framework to identify potential moderators that could influence the success of training programs. These included factors such as the grade level or age of participants, the setting of the intervention (e.g., school or workplace), the qualifications of the instructors, whether the training was universally applied or targeted specific groups, the use of technology, the duration of the intervention, and the time frame for evaluating outcomes. By exclusively focusing on meta-analyses, combining targeted search strategies, and employing detailed coding rules, our methodology ensures a robust and comprehensive assessment of how training programs impact the development of non-cognitive skills associated with the Big Five Personality Traits.

A.4 Additional Tables and Figures

Table A1: Filtered List of Literature in Our Non-Cognitive Skills Database (Final Papers)

No.	Author	Location	Income Group	Method	Type	Measure
1	Acosta et al. (2015)	Colombia	Developing	2	Big Five/Others	SD
2	Cunningham and Villaseñor (2016)	Peru	Developing	3	Big Five/Others	SD
3	Deming (2017)	USA	High income	3	Big Five/Others	SD
4	Díaz et al. (2012)	Peru	Developing	2,3	Big Five/Others	SD
5	Fletcher (2013)	USA	High income	2,3	Big Five/Others	Continuous
6	Glewwe et al. (2022)	China	Developing	3	Big Five/Others	SD
7	Heineck and Anger (2010)	Germany	High income	2,3	Big Five/Others	Dummy
8	Heineck (2011)	UK	High income	2,3	Big Five/Others	Dummy
9	Hilger et al. (2022)	Bangladesh	Developing	2,3	Big Five/Others	Continuous
10	Laible and Brenzel (2022)	Germany	High income	3	Big Five/Others	SD
11	Lavado et al. (2022)	Peru	Developing	3	Big Five/Others	Continuous
12	Lindqvist and Vestman (2011)	Sweden	High income	2	Big Five/Others	SD
13	McLean et al. (2019)	Canada	High income	3	Big Five/Others	SD
14	Otchia and Yamada (2019)	Ghana	Developing	2	Big Five/Others	SD
15	Palczyńska (2021)	Poland	High income	3	Big Five/Others	SD
16	Rozhkova et al. (2021)	Russia	Developing	3	Big Five/Others	SD

Table A2: Overview of Meta-Analyses on Non-Cognitive Skills Training

No.	Author	Type	Specific skill	School level*
1	Torgal et al. (2021)	Agreeableness	Empathy	ES, HS
2	Polanin et al. (2012)	Agreeableness	Empathy	ES, HS
3	Dignath et al. (2008)	Conscientiousness	Self-regulation	ES
4	Epton et al. (2017)	Conscientiousness	Goal-setting	Not mentioned
5	Frieese et al. (2017)	Conscientiousness	Self-control	Out of school
6	Li et al. (2021)	Conscientiousness	Self-control	PE, ES, HS
7	Piquero et al. (2016)	Conscientiousness	Self-control	PE, ES
8	Theobald (2021)	Conscientiousness	Self-regulation	PS
9	Dawson et al. (2020)	Emotional Stability	Mindfulness training	PS
10	Howard and Gutworth (2020)	Emotional Stability	Social skills	NA
11	Kraag et al. (2006)	Emotional Stability	Coping skills	ES
12	Moltrecht et al. (2021)	Emotional Stability	Emotional regulation	ES, HS
13	Kermani et al. (2022)	Extraversion	Communication skills	PS
14	Luo et al. (2022)	Extraversion	Socio-emotional competence	PE
15	McEwan et al. (2017)	Extraversion	Teamwork	NA
16	Murano et al. (2020)	Extraversion	Social and communication skills	PE
17	Shin and Lee (2021)	Extraversion	Pro-social behavior	ES, HS
18	Zhang et al. (2024)	Extraversion	Social interaction skills	PE, ES, HS
19	de Mooij et al. (2020)	Multiple	Inter-personal and emotional skills	ES, HS
20	Ang et al. (2022)	Openness	Resilience	PS
21	Bi et al. (2020)	Openness	Scientific creativity	PE, ES, HS, PS
22	Calo et al. (2024)	Openness	Resilience	PS
23	Ejaz et al. (2024)	Openness	Resilience	PS
24	Huang (2005)	Openness	Creativity	ES, HS
25	Kwon et al. (2016)	Openness	Creativity	ES, HS
26	Leppin et al. (2014)	Openness	Resilience	PS
27	Ma (2006)	Openness	Different skills	HS, PS
28	Sio and Lortie-Forgues (2024)	Openness	Creativity	ES, PS
29	Wang et al. (2024)	Openness	Creativity	PE, ES, HS, PS

* Note: Abbreviations: PE = pre-elementary, ES = elementary; HS = high school; PS = post-secondary.

Table A3: Descriptive Statistics, Labor Returns Analysis by Region

	Overall	Africa	Asia	Europe	Latin America	North America
Number of estimates	613	19	88	337	107	62
Estimate type (%)						
Big Five	92.7	79.0	96.6	95.2	90.7	80.6
Cognitive	7.3	21.0	3.4	4.8	9.3	19.4
Education control (%)						
Yes	47.0	52.6	98.9	33.8	45.8	45.2
No	53.0	47.4	1.1	66.2	54.2	54.8
Gender (%)						
Female	22.7	0.0	0.0	39.7	4.7	0.0
Male	22.8	0.0	0.0	40.1	4.7	0.0
Mixed	54.5	100.0	100.0	20.2	90.6	100.0
Methodology (%)						
OLS	28.6	0.0	2.3	41.0	23.4	16.1
IV	71.4	100.0	97.7	59.0	76.6	83.9

Figure A6: Big Five Personality Traits Questionnaire

How I am in general

Here are a number of characteristics that may or may not apply to you. For example, do you agree that you are someone who *likes to spend time with others*? Please write a number next to each statement to indicate the extent to which you agree or disagree with that statement.

1 Disagree Strongly	2 Disagree a little	3 Neither agree nor disagree	4 Agree a little	5 Agree strongly
---------------------------	---------------------------	------------------------------------	------------------------	------------------------

I am someone who...

- | | |
|--|--|
| 1. ____ Is talkative | 23. ____ Tends to be lazy |
| 2. ____ Tends to find fault with others | 24. ____ Is emotionally stable, not easily upset |
| 3. ____ Does a thorough job | 25. ____ Is inventive |
| 4. ____ Is depressed, blue | 26. ____ Has an assertive personality |
| 5. ____ Is original, comes up with new ideas | 27. ____ Can be cold and aloof |
| 6. ____ Is reserved | 28. ____ Perseveres until the task is finished |
| 7. ____ Is helpful and unselfish with others | 29. ____ Can be moody |
| 8. ____ Can be somewhat careless | 30. ____ Values artistic, aesthetic experiences |

Figure A7: The Wage Returns to Big Five Personality Traits, RE, Literature Main Results

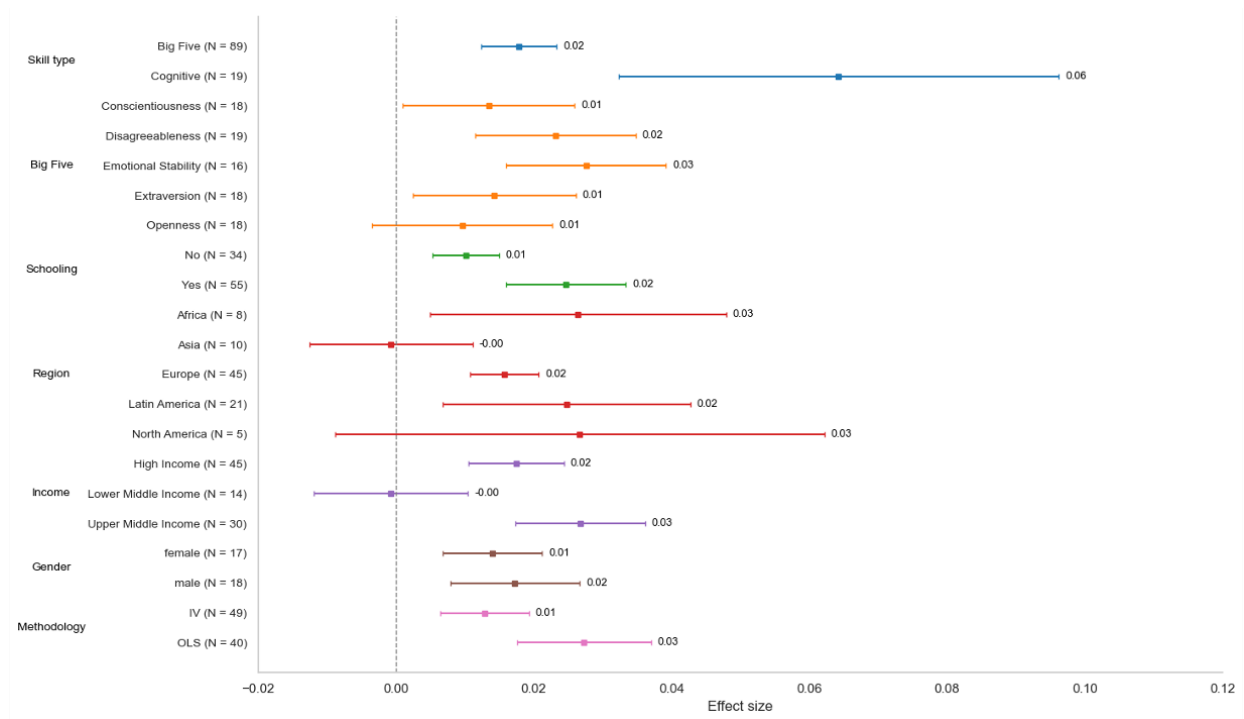


Figure A8: The Wage Returns to Big Five Personality Traits, FE, All Sample

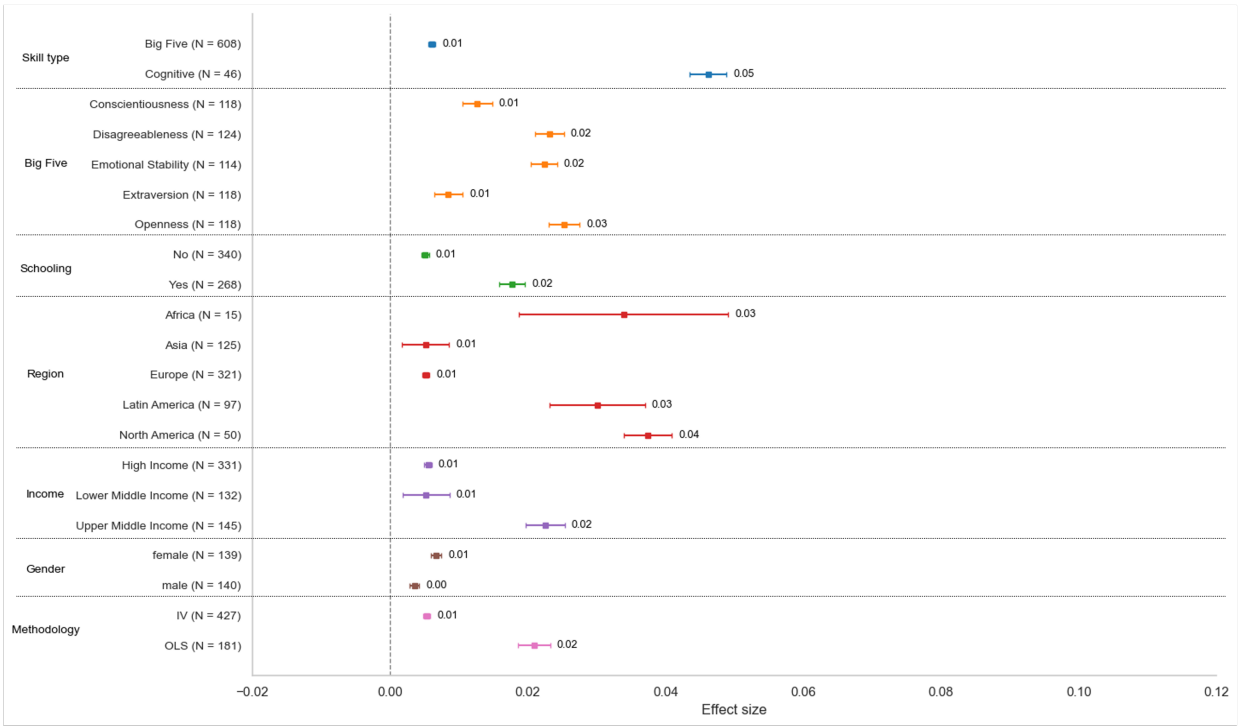


Figure A9: The Wage Returns to Big Five Personality Traits, RE, Only Papers Controlling for All Big Five Personality Traits

